February 25, 2025

Therapanacea SAS
Bhairavi Ajachandra
Compliance Director
7 bis boulevard Bourdon
Paris, 75004
France

Re: K242822
Trade/Device Name: ART-Plan+ (v.3.0.0)
Regulation Number: 21 CFR 892.5050
Regulation Name: Medical Charged-Particle Radiation Therapy System
Regulatory Class: Class II
Product Code: MUJ, QKB, LLZ
Dated: September 18, 2024
Received: September 18, 2024

Dear Bhairavi Ajachandra:

We have reviewed your section 510(k) premarket notification of intent to market the device referenced above and have determined the device is substantially equivalent (for the indications for use stated in the enclosure) to legally marketed predicate devices marketed in interstate commerce prior to May 28, 1976, the enactment date of the Medical Device Amendments, or to devices that have been reclassified in accordance with the provisions of the Federal Food, Drug, and Cosmetic Act (the Act) that do not require approval of a premarket approval application (PMA). You may, therefore, market the device, subject to the general controls provisions of the Act. Although this letter refers to your product as a device, please be aware that some cleared products may instead be combination products. The 510(k) Premarket Notification Database available at https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm identifies combination product submissions. The general controls provisions of the Act include requirements for annual registration, listing of devices, good manufacturing practice, labeling, and prohibitions against misbranding and adulteration. Please note: CDRH does not evaluate information related to contract liability warranties. We remind you, however, that device labeling must be truthful and not misleading.

If your device is classified (see above) into either class II (Special Controls) or class III (PMA), it may be subject to additional controls. Existing major regulations affecting your device can be found in the Code of Federal Regulations, Title 21, Parts 800 to 898. In addition, FDA may publish further announcements concerning your device in the Federal Register.

Additional information about changes that may require a new premarket notification are provided in the FDA guidance documents entitled "Deciding When to Submit a 510(k) for a Change to an Existing Device" (https://www.fda.gov/media/99812/download) and "Deciding When to Submit a 510(k) for a Software Change to an Existing Device" (https://www.fda.gov/media/99785/download).

Your device is also subject to, among other requirements, the Quality System (QS) regulation (21 CFR Part 820), which includes, but is not limited to, 21 CFR 820.30, Design controls; 21 CFR 820.90, Nonconforming product; and 21 CFR 820.100, Corrective and preventive action. Please note that regardless of whether a change requires premarket review, the QS regulation requires device manufacturers to review and approve changes to device design and production (21 CFR 820.30 and 21 CFR 820.70) and document changes and approvals in the device master record (21 CFR 820.181).

Please be advised that FDA's issuance of a substantial equivalence determination does not mean that FDA has made a determination that your device complies with other requirements of the Act or any Federal statutes and regulations administered by other Federal agencies. You must comply with all the Act's requirements, including, but not limited to: registration and listing (21 CFR Part 807); labeling (21 CFR Part 801); medical device reporting (reporting of medical device-related adverse events) (21 CFR Part 803) for devices or postmarketing safety reporting (21 CFR Part 4, Subpart B) for combination products (see https://www.fda.gov/combination-products/guidance-regulatory-information/postmarketing-safety-reporting-combination-products); good manufacturing practice requirements as set forth in the quality systems (QS) regulation (21 CFR Part 820) for devices or current good manufacturing practices (21 CFR Part 4, Subpart A) for combination products; and, if applicable, the electronic product radiation control provisions (Sections 531-542 of the Act); 21 CFR Parts 1000-1050.

All medical devices, including Class I and unclassified devices and combination product device constituent parts are required to be in compliance with the final Unique Device Identification System rule ("UDI Rule"). The UDI Rule requires, among other things, that a device bear a unique device identifier (UDI) on its label and package (21 CFR 801.20(a)) unless an exception or alternative applies (21 CFR 801.20(b)) and that the dates on the device label be formatted in accordance with 21 CFR 801.18. The UDI Rule (21 CFR 830.300(a) and 830.320(b)) also requires that certain information be submitted to the Global Unique Device Identification Database (GUDID) (21 CFR Part 830 Subpart E). For additional information on these requirements, please see the UDI System webpage at https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/unique-device-identification-system-udi-system.

Also, please note the regulation entitled, "Misbranding by reference to premarket notification" (21 CFR 807.97). For questions regarding the reporting of adverse events under the MDR regulation (21 CFR Part 803), please go to https://www.fda.gov/medical-devices/medical-device-safety/medical-device-reporting-mdr-how-report-medical-device-problems.

For comprehensive regulatory information about medical devices and radiation-emitting products, including information about labeling regulations, please see Device Advice (https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance) and CDRH Learn (https://www.fda.gov/training-and-continuing-education/cdrh-learn). Additionally, you may contact the Division of Industry and Consumer Education (DICE) to ask a question about a specific regulatory topic. See the DICE website (https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/contact-us-division-industry-and-consumer-education-dice) for more information or contact DICE by email (DICE@fda.hhs.gov) or phone (1-800-638-2041 or 301-796-7100).

Sincerely,

Lora D. Weidner, Ph.D.
Assistant Director
Radiation Therapy Team
DHT8C: Division of Radiological
Imaging and Radiation Therapy Devices
OHT8: Office of Radiological Health
Office of Product Evaluation and Quality
Center for Devices and Radiological Health

Enclosure

DEPARTMENT OF HEALTH AND HUMAN SERVICES
Food and Drug Administration

Indications for Use

Form Approved: OMB No. 0910-0120
Expiration Date: 07/31/2026
See PRA Statement below.

Submission Number *(if known)*

K242822

Device Name

ART-Plan+ (v.3.0.0)

Indications for Use *(Describe)*

ART-Plan+'s indicated target population is cancer patients for whom radiotherapy treatment has been prescribed. In this population, any patient for whom relevant modality imaging data is available.

ART-Plan+'s includes several modules:

- SmartPlan which allows automatic generation of radiotherapy treatment plan that the users import into their own Treatment Planning System (TPS) for the dose calculation, review and approval. This module is available for supported prescriptions for prostate only.

- Annotate which allows automatic generation of contours for organs at risk, lymph nodes and tumors, based on medical practices, on medical images such as CT and MR images

ART-Plan+ is not intended to be used for patients less than 18 years of age.

The indicated users are trained medical professionals including, but not limited to, radiotherapists, radiation oncologists, medical physicists, dosimetrists and medical professionals involved in the radiation therapy process.

The indicated use environments include, but are not limited to, hospitals, clinics and any health facility offering radiation therapy.

Type of Use *(Select one or both, as applicable)*

☒ Prescription Use (Part 21 CFR 801 Subpart D) ☐ Over-The-Counter Use (21 CFR 801 Subpart C)

CONTINUE ON A SEPARATE PAGE IF NEEDED.

K242822

510(k) Summary

This 510(k) Summary is submitted in accordance with 21 CFR Part 807, Section 807.92.

Contact details

Applicant Name:

TheraPanacea SAS

Applicant Address:

7 bis boulevard Bourdon
75004 Paris
France

Applicant Contact Telephone:

+33620604982

Applicant Contact:

Mrs. Bhairavi AJACHANDRA

Applicant Contact Email:

b.ajachandra@therapanacea.eu

Device Name:

Device Trade Name	Regulation Number	Common name	Device Class	Product Code(s)	Classification Name
ART-Plan+ (v3.0.0)	892.5050	Medical charged-particle radiation therapy system	Class II	MUJ Associated Product Code(s): QKB, LLZ	System, Planning, Radiation Therapy Treatment

Legally marketed predicate device

Predicate #	Predicate trade name (primary predicate is listed first)	Product code
K222728	Radiation Planning Assistant (RPA)	MUJ

Legally marketed reference devices

Reference device #	**Reference device trade name**	**Product code**
K213628	VBrain	QKB
K234068	ART-Plan	MUJ

Device Description Summary:

ART-Plan+ is a software platform allowing contour regions of interest on 3D images and to provide an automatic treatment plan. It includes several modules:

-Home: tasks

-Annotate and TumorBox: contouring of regions of interest

-SmartPlan: creation of an automatic treatment plan based on a planning CT and a RTSS

-Administration and settings : preferences management, user account management, etc.

-Institute Management: institute information, including licenses, list of users, etc.

-About: information about the software and its use, as well as contact details.

Annotate, TumorBox and SmartPlan are partially based on a batch mode, which allows the user to launch the operations of autocontouring and autoplanning without having to use the interface or the viewers. In that way, the software is completely integrated into the radiotherapy workflow and offer to the user a maximum of flexibility.

ART-Plan+ offers deep-learning based automatic segmentation of OARs and LNs for the following localizations:

-Head and neck (on CT images)

-Thorax/breast (on CT images)

-Abdomen (on CT and male on MR images)

-Pelvis male (on CT and MR images)

-Pelvis female (on CT images)

-Brain (on CT images and MR images)

ART-Plan+ offers deep-learning based automatic segmentation of targets for the following localizations:

-Brain (on MR images)

Intended Use/Indication for use:

ART-Plan+'s indicated target population is cancer patients for whom radiotherapy treatment has been prescribed. In this population, any patient for whom relevant modality imaging data is available.

ART-Plan+'s includes several modules:

- SmartPlan which allows automatic generation of radiotherapy treatment plan that the users import into their own Treatment Planning System (TPS) for the dose calculation, review and approval. This module is available for supported prescriptions for prostate only.

- Annotate which allows automatic generation of contours for organs at risk, lymph nodes and tumors, based on medical practices, on medical images such as CT and MR images

ART-Plan+ is not intended to be used for patients less than 18 years of age.

The indicated users are trained medical professionals including, but not limited to, radiotherapists, radiation oncologists, medical physicists, dosimetrists and medical professionals involved in the radiation therapy process.

The indicated use environments include, but are not limited to, hospitals, clinics and any health facility offering radiation therapy.

Indication for use comparison:

The intended use and indications for use of the proposed device, ART-Plan+ v3.0.0 and the primary predicate are similar as they are both softwares intended to be used in the planning of radiotherapy treatment:

-They allow creation of contours
-They allow the planification of external beam irradiation with photon beams using computerized tomography (CT) images.
-They allow the user to import the generated plan into their own Treatment Planning System (TPS) for dose calculation, review and approval.

The intended use and indications for use of the proposed device, ART-Plan+ v3.0.0 and the reference devices are similar as they are also softwares intended to be used in the planning of radiotherapy treatment in order to provide AI based segmentation tools to contour known (diagnosed) tumors, for organs at risk and lymph nodes.

Technological Comparison:

ART-Plan+ is substantially equivalent to the The Radiation Planning Assistant (RPA) (K222728) predicate device in the following aspects:

- Both devices allow the planification of external beam irradiation with photon beams using computerized tomography (CT) images.

- Both devices allow the creation of contours and treatment plans that the user imports into their own Treatment Planning System (TPS) for the dose calculation, review and approval.

There are no differences between the proposed device and the predicate that represent an additional claim for the proposed device.

Technical Information			
Property	***Proposed Device*** **ART-Plan v3.0.0**	***Primary Predicate*** **Radiation Planning System (RPA)**	**Comment**
Delineation Method	AI	AI	The proposed device and primary predicate share an AI delineation method.
Segmentation Features	Automatically delineates OARs and lymph nodes and targets. Deep learning algorithm. Automatic segmentation includes the following localizations: * head and neck (on CT) * thorax/breast (for male/female and on CT) * abdomen (on CT images and MR images) * pelvis male (on CT and MR) * pelvis female (on CT images) * brain (on CT images and MR images)	The RPA provides autocontouring for a range of structures	The proposed device proposes segmentation on the same localizations with the same medical images of different modalities using AI.
Regions and Volumes of Interest (ROI)	AI Based autocontouring	AI Based autocontouring,	The proposed device and the primary predicate allow AI automatic contouring.

Non-Clinical and/or Clinical Tests Summary & Conclusions

In order to determine the substantial equivalence of ART-Plan + v3.0.0, the following tests have been performed:

Non regression testing of autosegmentation of organs at risk - Validation of the performance of autosegmentation (on CT and MR images) of already existing structures after retraining or algorithm improvement was performed by comparison of the DSC between contours generated by the previous version of ART-Plan (v2.2.0) and manual contours performed by medical experts and the DSC between contours generated by the new version of ART-Plan + (v3.0.0) and manual contours performed by medical experts. The evaluation was performed on a minimum sample size of 24 patients.

The contours are considered acceptable for clinical use if the following acceptance criterion is achieved:

-Mean DSC should not regress negatively between the current and last validated version of Annotate beyond a maximum tolerance margin set to -5% relative error.

Qualitative evaluation of autosegmentation of organs at risk - Validation of the performance of autosegmentation of new organs at risk (or organs at risk not passing the non regression testing) on CT images was performed by qualitative evaluation of the contours by medical experts. This evaluation was performed on a minimum sample size of 18 patients.

The contours are considered acceptable for clinical use if the following acceptance criterion is achieved:

-The clinicians' qualitative evaluation of the auto-segmentation is considered acceptable for clinical use without modifications (A) or with minor modifications / corrections (B) with a A+B % above or equal to 85% considering the following scale:

A. the contour is acceptable for a clinical use without any modification

B. the contour would be acceptable for clinical use after minor modifications/corrections

C. the contour requires major modifications (e.g. it would be faster for the expert to manually delineate the structure)"

Quantitative evaluation of autosegmentation of organs at risk – validation of the performance of autosegmentation of new organs at risk on CT images was performed by quantitative evaluation of the contours. This evaluation was performed on a minimum sample size of 24 patients.

The contours are considered acceptable for clinical use if the following acceptance criterion is achieved:

- Mean DSC (annotate) ≥ 0.8

Inter-expert variability evaluation of autosegmentation of organs at risk – validation of the performance of autosegmentation of new organs at risk on CT images was performed by an inter-expert variability evaluation of the contours. The same images were asked to be contoured by two independent medical experts. DSC was calculated between them and with contours generated by the software and the compared. This evaluation was performed on a minimum sample size of 13 patients.

The contours are considered acceptable for clinical use if the following acceptance criterion is achieved:

- Mean DSC (annotate) ≥ Mean DSC (inter-expert) with a tolerance margin of –5% of relative error

Quantitative evaluation of autosegmentation of Brain metastasis – Validation of the performance of autosegmentation of brain metastasis on MR images was performed by quantitative evaluation. Contours provided by medical experts were compared with contours generated by the software. This evaluation was performed on a minimum sample size of 51 tients.

The contours are considered acceptable for clinical use if the following acceptance criterion is achieved:

- The lesion-wise sensitivity is equal to or superior to state-of-the-art as benchmark: mean lesion-wise sensitivity ≥ 0.86

AND

- The lesion-wise precision is equal to or superior to state-of-the-art as benchmark: mean lesion-wise precision ≥ 0.70

AND

- The lesion-wise DSC similarity coefficient (DSC) is equal to or superior to state-of-the-art as benchmark: mean lesion-wise DSC ≥ 0.78

AND

- The patient-wise DSC similarity coefficient (DSC) is equal to or superior to state-of-the-art as benchmark: mean patient-wise DSC ≥ 0.83

AND

- The patient-wise false positive (FP) is equal to or superior to state-of-the-art as benchmark: mean patient-wise FP ≤ 2.1

Quantitative evaluation of autosegmentation of Glioblastoma – Validation of the performance of autosegmentation of glioblastoma on MR images was performed by quantitative evaluation. Contours provided by medical experts were compared with contours generated by the software. This evaluation was performed on a minimum sample size of 43 patients.

The contours are considered acceptable for clinical use if the following acceptance criterion is achieved:

-The sensitivity is equal to or superior to state-of-the-art as benchmark: mean sensitivity ≥ 0.80

AND

-The DSC is equal to or superior to state-of-the-art as benchmark: mean DSC ≥ 0.76

Quantitative and qualitative evaluation of automatic treatment plans generations – validation of the performance of automatic radiotherapy treatment plans generation was performed by both quantitative and qualitative evaluation. The quantitative evaluation consisted of direct comparison with manual plans. The qualitative evaluation consisted of medical experts determining the clinical acceptability of plans. These evaluations were performed on a minimum sample size of 20 patients.

- The treatments plans are considered acceptable for clinical use if the following acceptance criteria are achieved:

Quantitative evaluation: effectiveness difference (%) in DVH achieved goals between manual plans and automatic plans ≤ 5%

AND

Qualitative evaluation: % of clinical acceptable automatic plans ≥ 93% after expert review.

All validation tests were carried out using datasets representative of the worldwide population receiving radiotherapy treatments. Finally, all tests passed their respective acceptance criteria, thus showing ART-Plan + v3.0.0 clinical acceptability.

Not applicable

Based on the information presented in these 510(k) premarket notifications the TheraPanacea ART-Plan is considered substantially equivalent. The TheraPanacea ART-Plan + 3.0.0 is as safe and effective as the currently marketed predicate device.

Based on testing and comparison with the predicate device, TheraPanacea ART-Plan indicated no adverse indications or results. It is our determination that the TheraPanacea ART-Plan + 3.0.0 performs within its design specifications and is substantially equivalent to the predicate.